

10.4 - Consumer Debt

Problem Solving, Math 1000

Bryce A Christopherson

1 Key Idea: Credit Cards

Consumer debt is, by itself, certainly not a bad thing. Credit cards make modern life significantly easier, and if a person uses one wisely, they often carry with them no cost and mild rewards. However, if a person fails to pay the balance due in full before the payment date (we call this carrying a revolving balance), you will almost always accrue finance charges on the balance to be paid during the next billing cycle. And this is only if you fail to pay the whole thing off – if you fail to pay at all, often you will also receive a late payment penalty on top of the finance charges.

So, with this in mind, it is definitely a good thing to understand how credit cards work and how finance charges are calculated on a credit card statement.

1.1 Example

Let's say you have a MasterCard with APR of 17.2% on standard purchases. The billing cycle starts on the 25th of each month and ends on the 24th of the next month. Let's also suppose that you are normally very good about paying the balance due in full each month, but April was especially expensive and you did not pay off your balance due entirely. For the April 15 - May 24 billing cycle, you carried over a \$500 balance to the next cycle. When your May 25 - June 24 credit card bill arrives, these are the charges:

- May 25: Previous balance (\$500)
- May 28: Phone bill payment (\$85.75)
- May 25: Gasoline purchase (\$45)
- May 25: Groceries (\$73.60)

Since you didn't pay your previous balance in full, there will also be a finance charge in your bill. With most credit cards, finance charges are computed using simple interest applied to the average daily balance for the billing cycle. So, there are two things to calculate: The periodic interest rate and the average daily balance.

The periodic interest rate is found by taking the APR and dividing by 365, then multiplying by the number of days in the billing cycle. In our example (since May 25 - June 24 is 31 days), this is then

$$p = \frac{(0.172)(31)}{365} \approx 0.01464$$

The average daily balance is found by adding the daily balance for each day in the billing cycle and dividing by the total number of days. This seems like a lot of work (and definitely can be!), but in our example, this is relatively simple: all we need do is look at the length of the periods where our balance remained constant. This is summarized as follows:

- May 25 - May 27: \$500.00 (3 days)
- May 28 - June 4: \$585.75 (8 days)
- June 5 - June 15: \$630.75 (11 days)
- June 16 - June 24: \$704.35 (9 days)

This allows us to calculate the average daily balance as follows:

$$ADB = \frac{(3)(500) + (8)(585.75) + (11)(630.75) + (9)(704.35)}{31} = \$627.85$$

Now, having done this, we can calculate our finance charge. Since credit card interest is simple over a billing cycle, we just multiply the average daily balance by the periodic interest rate. In this case, we get

$$\text{finance charge} = (627.85)(0.01464) = \$9.19$$

So, with this, we can expect that the new balance on our credit card bill next month will be

$$\begin{aligned} \text{new balance} &= \text{previous balance} + \text{purchases} + \text{finance charge} \\ &= \$500 + \$204.35 + \$9.19 \\ &= \$713.54 \end{aligned}$$

If you pay this off at the end of the month, you're good. If not, you can expect to see interest charges continue to accumulate on all your future credit card purchases.

1.2 Key Idea: Installment Loans

Credit card debt is often not intentional. However, we often do take on debt quite intentionally, and generally in the form of an installment loan. Installment loans are probably the most familiar loans to most of us: The borrower pays back the loan plus interest in equal installments paid at regular intervals over the life of the loan. This process of paying off a loan by making regular payments is called amortization, and we often say a loan is being amortized over a certain length of time.

In the US, almost all installment loans are based on monthly payments, and generally the interest is compounded on a monthly basis as well. The following formula, known as the amortization formula, gives the monthly payment M on a loan with principle P and APR r which is to be paid out over T monthly installments:

$$M = P \frac{p(1+p)^T}{(1+p)^T - 1}, \text{ where } p = \frac{r}{12} \text{ is the monthly interest rate}$$

1.3 Example

Let's say you're buying a new car. Suppose you found a great car and negotiate a good price with the dealer: \$23,995 including taxes and license fees. In addition, suppose as well that the car manufacturer is offering a promotion where you can choose between a \$2,000 rebate or special financing at 1.2% APR for 60 months. If you choose the manufacturer's rebate, then you have to finance the balance at the dealer's rate of 6.48% APR for a 60-month loan. So, which one do you choose?

If you take the \$2,000 manufacturer's rebate and finance the balance through the dealer, you will have a principal of $P = 21995$, an APR of $r = 0.0648$, a monthly interest rate of $p = \frac{0.0648}{12} = 0.0054$, and a total of $T = 60$ monthly payments. Plugging this into the amortization formula gives monthly payments of:

$$M = (21995) \frac{(0.0054)(1.0054)^{60}}{(1.0054)^{60} - 1} = \$430.15$$

If you choose the special financing offer instead, you then have $P = \$23,995$, an APR $r = 0.012$, a monthly interest rate of $p = \frac{0.012}{12} = 0.001$, and a total of $T = 60$ monthly payments. The amortization formula gives monthly payments of

$$M = (23995) \frac{(0.001)(1.001)^{60}}{(1.001)^{60} - 1} = \$412.23$$

So, at the end of the day, you can see that the special financing offer is definitely better than the manufacturer's rebate. Over the life of the loan, you'd save over \$1,000 by choosing this option.